MTH636 – Game Theory

Lecture 7: Nash Equilibrium

1 Formal Definition of Nash Equilibrium

Definition: Nash Equilibrium

For a game $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, a strategy profile s^* is a **Nash Equilibrium (NE)** if for all $i \in N$ and for all $s_i \in S_i$:

$$u_i(s^*) \geq u_i(s_i, s^*_{-i})$$

In words: No player i can strictly increase their payoff by unilaterally changing their strategy from s^*_i to some other s_i, given that all other players play their part of s^*_{-i}.

Example: Example: Battle of the Sexes (Variants)

Consider the following game matrix:

$P_1 \setminus P_2$	F	D
F	(2, 1)	(0, 0)
D	(0, 0)	(1, 2)

Here, both (F, F) and (D, D) are Nash Equilibria. If they are at (F, F), neither wants to deviate to D (yielding 0).

2 Best Reply Correspondence

To find Nash Equilibria, it is often useful to look at players' best responses.

Definition: Best Reply Correspondence

The best reply correspondence for player i is a function $b_i : S_{-i} \to 2^{S_i}$ (mapping to the power set of S_i), defined as:

$$b_i(s_{-i}) = \arg\max_{t_i \in S_i} u_i(t_i, s_{-i})$$

This gives the set of all strategies t_i that maximize player i's payoff given the opponents' profile s_{-i}.

Theorem: Nash Equilibrium in terms of Best Replies

A strategy profile s^* is a Nash Equilibrium if and only if:

$$s^*_i \in b_i(s^*_{-i}) \quad \forall i \in N$$

3 Finding Nash Equilibria in Specific Games

3.1 Beauty Contest Game

Recall the Beauty Contest Game: $N = \{1, 2, 3\}$, $S_i = [0, 100]$. Target $T(x_1, x_2, x_3) = \frac{2}{3}\left(\frac{x_1+x_2+x_3}{3}\right)$. Players who are closest to T win.

Is the profile $(0, 0, 0)$ a Nash Equilibrium? Yes, because if everyone plays 0, the target is 0, and deviating yields a strictly positive number which is further from 0.

What about profiles of the form (a, a, a) where $a \in (0, 100]$? These are **not** Nash Equilibria, because any player could deviate to a slightly lower number to get closer to $\frac{2}{3}$ of the average and win outright.

3.2 Second-Price Auction

Recall that bidding true valuation weakly dominates all other strategies. Therefore:

- The profile $(v_1, v_2, \ldots, v_n)$ is a Nash Equilibrium.
- What about $(v_1, 0, 0, \ldots, 0)$? If v_1 is the highest valuation, Player 1 wins and pays 0. No other player can profitably deviate (bidding higher than v_1 means they win but pay more than their valuation, giving a negative payoff). So this can also be an NE under certain conditions.

Exercise: Find all Nash Equilibria of the Second-Price Auction game where $v_1 = 10, v_2 = 11$.

4 Continuous Action Spaces (Cournot-like Game)

Consider a 2-player game where $N = \{F_1, F_2\}$ and $S_i = [0, 1]$. Payoffs are given by:

$$u_i(a_1, a_2) = (2 - a_1 - a_2)a_i - c_i a_i$$

where c_1, c_2 are cost parameters.

To find the Nash Equilibrium (a_1^*, a_2^*), we use the Best Reply functions. Player 1 maximizes $u_1(a_1, \bar{a}_2)$ with respect to a_1:

$$\frac{\partial u_1}{\partial a_1} = 2 - 2a_1 - \bar{a}_2 - c_1 = 0 \implies a_1 = \frac{2 - c_1 - \bar{a}_2}{2}$$

Thus, the best reply function is $b_1(\bar{a}_2) = \frac{2-c_1-\bar{a}_2}{2}$. Similarly, for Player 2: $b_2(\bar{a}_1) = \frac{2-c_2-\bar{a}_1}{2}$.

At the Nash Equilibrium, both must be playing best responses to each other:

$$a_1^* = \frac{2 - c_1 - a_2^*}{2} \quad \text{and} \quad a_2^* = \frac{2 - c_2 - a_1^*}{2}$$

Substitute a_2^* into the equation for a_1^*:

$$\begin{aligned} a_1^* &= \frac{2 - c_1 - \left(\frac{2-c_2-a_1^*}{2}\right)}{2} \\ 2a_1^* &= 2 - c_1 - 1 + \frac{c_2}{2} + \frac{a_1^*}{2} \\ 4a_1^* &= 2 - 2c_1 + c_2 + a_1^* \\ 3a_1^* &= 2 - 2c_1 + c_2 \implies a_1^* = \frac{2 - 2c_1 + c_2}{3} \end{aligned}$$

By symmetry, $a_2^* = \frac{2-2c_2+c_1}{3}$.

The equilibrium payoffs are:

$$u_1(a_1^*, a_2^*) = (a_1^*)^2 \quad \text{and} \quad u_2(a_1^*, a_2^*) = (a_2^*)^2$$